Hamiltonian simulation in the low energy subspace

Quantum Algorithms, Spring 2026 Project Report

Shradha Santosh Nair
PhD, Quantum Information Theory
IIIT Hyderabad
shradha.nair@research.iiit.ac.in
Roll No. 2025811001

Harsh Kapoor
Dual Degree, Electronics and Communication
IIIT Hyderabad
harsh.kapoor@research.iiit.ac.in
Roll No. 2023112004

Abstract

Quantum simulation algorithms often suffer from high complexity due to their reliance on the spectral norm of the entire Hamiltonian. However, physically relevant systems typically evolve within a low-energy subspace. This project investigates the profound algorithmic advantages of restricting Hamiltonian simulation to this subspace. We synthesize key results, from exponentially decaying bounds on high-energy leakage in standard Suzuki-Trotter formulas to their recent adaptations for time-dependent Hamiltonians. Furthermore, we establish connections between local energy distributions and the area law of entanglement. We also recreate the papers results of a 2x6 spin system and also present a 8 spin chain version extension with higher order approximations. By anchoring the simulation in the low-energy sector, product formulas become remarkably more efficient, advancing the feasibility of near-term quantum algorithms.

I. Introduction

The simulation of quantum dynamics is a flagship application for quantum computers, promising breakthroughs in many-body physics, quantum chemistry, and material science. Digital Hamiltonian simulation typically relies on product formulas, such as Suzuki-Trotter decompositions [1], to approximate the time-evolution operator. However, a major algorithmic bottleneck persists in this approach: the simulation error—and consequently the required quantum gate depth—scales proportionally to the global spectral norm of the full Hamiltonian. For large, highly interactive systems, this norm can be overwhelmingly large or unbounded, rendering simulations prohibitively expensive even if the system never actually visits high-energy states.

To resolve this limitation, recent theoretical paradigms have pivoted to simulating the physically relevant regime: the low-energy subspace. Şahinoğlu and Somma [2] established that if an initial quantum state is supported primarily within a low-energy sector, the overall simulation complexity drops significantly. By proving that the probability of the state leaking into high-energy subspaces decays exponentially with the energy gap, they showed that algorithmic complexity scales only with a localized, ”effective” low-energy norm. This fundamentally static framework was recently extended by Zhou et al. [3] to time-dependent Hamiltonians, proving that exponential reductions in system-size dependence can be maintained dynamically during non-equilibrium processes and adiabatic state preparation.

A critical physical property underpinning these low-energy dynamics is the area law of entanglement. Restricting a system to a low-energy subspace inherently limits local correlations. As demonstrated by Arad, Kuwahara, and Landau [4], global and local energy distributions in quantum spin lattices are deeply connected; truncated local Hamiltonians possess low-energy spectra that are exponentially close to the exact global Hamiltonian. This spectral locality serves as the mathematical bedrock for efficiently simulating 1D and 2D spin systems.

The remainder of this report is organized as follows: Section II introduces product formulas and the mechanics of Trotterization; Section III details the mathematical structure of local Hamiltonians; Section IV defines the high-energy leakage bounds; Section V derives the main low-energy error bounds; Section VI presents our numerical simulations and empirical observations for the 1D chain and 2×6 ladder systems; Section VII discusses the physical relevance of the positivity of local terms; and Section VIII concludes the report with a discussion of connections of this work to area laws.

II. Product Formulas and Suzuki–Trotter Approximations

A. Hamiltonian Decomposition and Time Evolution

Consider a time-independent Hamiltonian that can be expressed as a sum of L Hermitian operators:

$$H = \sum_{l=1}^{L} H_l. \tag{1}$$

The corresponding time-evolution operator is given by

$$U(t) = e^{-itH} = e^{-it\sum_{l=1}^{L} H_l}. \tag{2}$$

In general, directly implementing this unitary is difficult, as exponentiating a sum of non-commuting operators leads to high circuit complexity. To overcome this, we approximate the global evolution using *product formulas* (like Lie–Trotter or

Suzuki–Trotter approximations), which decompose the evolution into a sequence of exponentials of individual Hamiltonian terms over short time intervals. Each such exponential corresponds to evolution under a simpler operator that is typically much easier to implement on quantum hardware.

B. Product Formulas

Let $W_p(s)$ denote a product formula approximation of order p for the evolution operator $U(s)$, where $s \in \mathbb{R}$ represents a short time interval. The general form of such a product formula is

$$W_p(s) = e^{-is_q H_{l_q}} \cdots e^{-is_2 H_{l_2}} e^{-is_1 H_{l_1}}, \tag{3}$$

where each coefficient $s_j \in \mathbb{R}$ is proportional to s, and each index satisfies $1 \leq l_j \leq L$. Intuitively, this expression represents a sequence of short-time evolutions under individual Hamiltonian components.

We define the total effective evolution time of the product formula as

$$|s| = \sum_{j=1}^{q} |s_j|, \tag{4}$$

and assume that L and q are constants, i.e., $L, q = O(1)$.

a) p-th Order Product Formula: A product formula $W_p(s)$ is said to be *p-th order* if it approximates the exact evolution operator $U(s)$ with error scaling as

$$\|U(s) - W_p(s)\| = O\left((h|s|)^{p+1}\right), \tag{5}$$

where $h = \max_l \|H_l\|$.

An equivalent characterization can be obtained using the Baker–Campbell–Hausdorff (BCH) expansion. Under this expansion, the product formula $W_p(s)$ reproduces the exact evolution e^{-isH} up to all terms of order s^p, and the first non-vanishing correction appears at order s^{p+1}. These leading error terms are composed of nested commutators of the Hamiltonian components, with commutator depth $p+1$. The detailed mathematical derivation of this statement and the nested commutator scaling can be found in Appendix A and B.

Higher-order product formulas can be constructed systematically using Suzuki's recursive method, which cancels lower-order commutator contributions by symmetrizing and combining lower-order approximations.

C. Trotterization over Finite Time

To simulate the evolution over a finite time t, we divide the interval into r segments of equal length:

$$s = \frac{t}{r}. \tag{6}$$

The full evolution operator can then be approximated as

$$U(t) \approx (W_p(s))^r. \tag{7}$$

The integer r is referred to as the *Trotter number*, and it determines the trade-off between accuracy and computational cost. Increasing r reduces the time step s, thereby improving the approximation accuracy. If each exponential $e^{-is_j H_{l_j}}$ is itself decomposed into M elementary operations, then the total number of operations required is proportional to

$$qMr = O(Mr). \tag{8}$$

Thus, the overall simulation complexity is directly governed by the choice of the Trotter number r and the order p of the product formula.

D. Improved Error Bounds for Low-Energy States

The standard error bound depends on the quantity $h = \max_l \|H_l\|$, which captures the largest operator norm among the Hamiltonian components. In many physical systems, this quantity can be extremely large, leading to highly pessimistic error estimates. However, these bounds can be significantly improved when the initial state is restricted to a low-energy subspace.

Suppose the initial state satisfies

$$\Pi_{\leq \Delta} |\psi\rangle = |\psi\rangle, \tag{9}$$

where $\Pi_{\leq \Lambda}$ denotes the projector onto the subspace spanned by eigenstates of H with eigenvalues at most $\Lambda \geq 0$. We also assume that each H_l is positive semidefinite.

Under this assumption, the dynamics of the system are effectively confined to a subspace of limited energy, and therefore do not explore the full, potentially unbounded spectral range of the full Hamiltonian.

E. Effective Operators

For a Hermitian operator X and a cutoff $\Delta_0 \geq \Delta$, we define the corresponding effective operator as

$$\bar{X} = \Pi_{\leq \Delta_0} X \Pi_{\leq \Delta_0}. \tag{10}$$

This operator acts only within the truncated low-energy subspace and remains Hermitian. Using this definition, we construct the effective Hamiltonian

$$\bar{H} = \sum_l \bar{H}_l, \tag{11}$$

and the corresponding effective time-evolution operator

$$\bar{U}(s) = e^{-is\bar{H}}. \tag{12}$$

Similarly, we define $\bar{W}_p(s)$ by replacing each H_l with $\bar{H}_l$ in the product formula. Since the initial state lies entirely within the low-energy subspace, the evolution under the full Hamiltonian and the effective Hamiltonian coincide exactly:

$$U(t)|\psi\rangle = \bar{U}(t)|\psi\rangle. \tag{13}$$

We now define the effective norm bound

$$\bar{h} = \max_l \|\bar{H}_l\| \leq \Delta_0. \tag{14}$$

Applying the standard product formula error bound to the effective Hamiltonian yields

$$\|(U(s) - \bar{W}_p(s))|\psi\rangle\| = O\left((\Delta_0 s)^{p+1}\right). \tag{15}$$

This represents a massive improvement over the general worst-case bound when $\Delta_0 \ll h$, which is the case when the state is confined to a low-energy subspace. Although the effective product formula $\bar{W}_p(s)$ is not directly implementable in an algorithm, it can be proven that the original, implementable product formula $W_p(s)$ achieves this same improved error scaling when applied to such restricted states.

III. Local Hamiltonians

We consider the problem of simulating the time evolution of a quantum system composed of N spins arranged on a lattice. The dynamics of the system are governed by a Hamiltonian H, and the corresponding time-evolution operator is given by

$$U(t) = e^{-iHt}. \tag{16}$$

Directly simulating this exponential is generally intractable for large systems, and therefore it is essential to exploit the structure of physically relevant Hamiltonians.

A. Local Hamiltonians

A Hamiltonian is said to be *k-local* if it can be expressed as a sum of terms, each acting non-trivially on at most k spins, where k is a constant independent of the system size N. Thus, we write

$$H = \sum_\alpha h_\alpha, \tag{17}$$

where each operator h_α acts on at most k spins.

In addition, we assume that each local interaction term has bounded strength,

$$\|h_\alpha\| \leq J, \tag{18}$$

where J sets the characteristic energy scale of the system.

Another important parameter is the *degree* d of the Hamiltonian, defined as the maximum number of local interaction terms that involve any given spin. This ensures that no spin participates in an unbounded number of interactions. As a consequence, the total number of local terms scales at most linearly with system size,

$$\text{number of terms} \leq dN. \tag{19}$$

These assumptions capture a broad class of physically relevant systems, including spin chains, lattice models, and systems with both short-range and long-range interactions.

B. Decomposition into Commuting Layers

To efficiently simulate the time evolution, it is useful to reorganize the Hamiltonian into a sum of groups of commuting terms. Specifically, we write

$$H = \sum_{l=1}^{L} H_l, \tag{20}$$

where each H_l is itself a sum of M local terms,

$$H_l = \sum_{j=1}^{M} h_{l,j}. \tag{21}$$

The key property of this decomposition is that all terms within a given group commute,

$$[h_{l,i}, h_{l,j}] = 0 \quad \forall i, j, \tag{22}$$

while terms belonging to different groups may not commute. This structure can be obtained, for example, by graph-coloring arguments, where interaction terms are assigned to different groups such that no two overlapping terms appear in the same group.

Since each spin is involved in at most d interaction terms, one can ensure that the total number of such groups L remains bounded (i.e., $L = O(1)$ for constant d and k). Furthermore, the total number of local terms satisfies

$$LM \leq dN. \tag{23}$$

C. Commuting vs Non-Commuting Contributions

The distinction between commuting and non-commuting terms is crucial for simulation. If two operators commute, their exponentials factorize exactly:

$$e^{-i(A+B)t} = e^{-iAt}e^{-iBt}, \quad \text{if } [A, B] = 0. \tag{24}$$

Therefore, for each group H_l composed of commuting terms, we have an exact decomposition:

$$e^{-iH_l t} = \prod_{j=1}^{M} e^{-ih_{l,j}t}. \tag{25}$$

In contrast, operators belonging to different groups generally do not commute. This non-commutativity arises when interaction terms act on overlapping spins or involve non-commuting operators. For example, two terms acting on a shared spin with different Pauli operators will fail to commute. As a result,

$$e^{-i(H_1+H_2)t} \neq e^{-iH_1 t}e^{-iH_2 t} \tag{26}$$

in general.

D. Implications for Hamiltonian Simulation

This decomposition enables the use of product formulas (Trotter-Suzuki approximations) to simulate the full time evolution:

$$e^{-iHt} \approx \left(\prod_{l=1}^{L} e^{-iH_l s} \right)^r, \tag{27}$$

where $t = rs$ and s is a small time step.

The error in this approximation arises solely from the non-commutativity between different groups H_l, while the decomposition within each group is exact. This structure is therefore essential for:

- Reducing the simulation of a many-body system to sequences of few-body evolutions,
- Enabling efficient implementation on quantum hardware,
- Providing controlled and analyzable approximation errors.

IV. Leakage to High-Energy States

We consider a Hamiltonian of the form $H = \sum_{l=1}^{L} H_l$, where each H_l is a local, positive semidefinite operator. To analyze the energy dynamics, we define two spectral projectors: $\Pi_{\leq\Delta}$, which projects onto the subspace spanned by eigenstates of H with energies $\leq \Delta$, and $\Pi_{>\Lambda_0}$, which projects onto the subspace spanned by eigenstates with energies $> \Lambda_0$. Here, Δ represents the maximum energy bound of the initial state, while Λ_0 is an intermediate energy threshold satisfying $\Lambda_0 \geq \Delta$. We operate under the assumption that the initial state is entirely contained within the low energy space, satisfying $\Pi_{\leq\Delta}|\psi\rangle = |\psi\rangle$. Our primary objective is to quantify how much amplitude leaks into higher-energy states under evolution driven by a single local term H_l.

A. Leakage Bound

We evaluate the norm of the transition operator $\|\Pi_{>\Lambda_0} e^{-isH_l} \Pi_{\leq\Delta}\|$, which strictly measures the extent to which a state, initially confined to energies $\leq \Delta$, transitions into the subspace of energies $> \Lambda_0$ after a short time evolution.

*a) Lemma (**Leakage Bound**):* For local Hamiltonians, the leakage is bounded by:

$$\|\Pi_{>\Lambda_0} e^{-isH_l} \Pi_{\leq\Delta}\| \leq e^{-\alpha_1(\Lambda_0-\Delta)/J} \left(e^{\alpha_2 J|s|N} - 1\right), \tag{28}$$

for strict positive constants $\alpha_1, \alpha_2 > 0$. This relationship is taken from a paper by Itai Arad et. al [4]. A complete step-by-step derivation of this leakage bound is provided in Appendix C.

We define J as a strict upper bound on the strength of local interaction terms in the Hamiltonian. If the Hamiltonian is written as a sum of local terms h_X, then $J = \max_X \|h_X\|$. This parameter establishes the natural energy scale of the quantum system and plays a central role in controlling both the growth of Trotter errors and the rate of leakage to higher-energy states.

B. Interpretation of the Bound

This inequality captures two competing physical effects during simulation. First, an energy gap suppression effect is governed by the factor $e^{-\alpha_1(\Lambda_0-\Delta)/J}$, demonstrating that transitions to higher-energy states are exponentially suppressed by the size of the energy gap itself. Second, a local spreading effect is captured by the factor $e^{\alpha_2 J|s|N}$, which reflects the physical reality that repeated application of local operators increases the spatial support and therefore the entanglement footprint of the state, allowing energy to spread across the lattice. Thus, although local evolution can temporarily generate higher-energy components, this effect is strongly suppressed when the energy gap $\Lambda_0 - \Delta$ is large and the local evolution time step $|s|$ is kept small.

C. Role in Hamiltonian Simulation

It theoretically guarantees that the simulated dynamics remain mostly localized within a low-energy subspace, and that any errors arising from rare transitions to higher-energy states can be strictly bounded and controlled. Consequently, we can analytically replace the full Hamiltonian with an effective low-energy Hamiltonian without introducing large, unbounded simulation errors. This forms the backbone for the improved error scaling presented in our Main Result.

V. Main Result: Low-Energy Error Bounds for Product Formulas

Here is the central result of this paper, which establishes that the error of product formulas depends on an *effective low-energy scale* rather than the full, potentially unbounded Hamiltonian norm [2].

A. Setup and Notation

We consider a Hamiltonian of the form $H = \sum_{l=1}^{L} H_l$, where each H_l is a local, positive semidefinite operator, and an initial state that satisfies $\Pi_{\leq\Delta}|\psi\rangle = |\psi\rangle$. To structure our bounds, we define Δ as the maximum intrinsic energy of the initial state, and $\Delta_0 \geq \Delta$ as an effective low-energy spectral cutoff. We let $U(s) = e^{-isH}$ denote the exact time-evolution operator, and $W_p(s)$ denote a p-th order product formula approximation.

B. Statement of the Result

*a) Theorem (**Low-Energy Product Formula Error**):* For a local Hamiltonian and a p-th order product formula $W_p(s)$, the approximation error satisfies:

$$\|(U(s) - W_p(s))\Pi_{\leq\Delta}\| = O\left((\Delta_0|s|)^{p+1}\right), \tag{29}$$

where the effective energy scale is bounded by:

$$\Delta_0 = \Delta + \beta_1 J \log\left(\frac{\beta_2}{J|s|}\right) + \beta_3 J^2 N|s|, \tag{30}$$

and $\beta_1, \beta_2, \beta_3$ are positive constants depending on the system geometry, and builds up on the result of the lemma 1.

C. Interpretation of the Result

This theorem definitively proves that the error of product formulas is governed by the effective energy scale Δ_0, rather than the worst-case operator norm $h = \max_l \|H_l\|$. The quantity Δ_0 is composed of three distinct physical contributions: the intrinsic energy of the initial state Δ; a mathematical correction $J\log(1/(J|s|))$ arising from balancing leakage errors; and the accumulated energy growth $J^2N|s|$ caused by local interactions spreading across the N-spin system over time s. Crucially, when the initial state lies deep in a low-energy subspace (i.e., $\Delta \ll h$), this error bound represents an improvement over standard worst-case bounds.

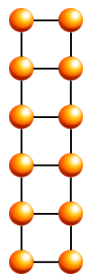
D. Key Idea Behind the Result

The mathematical proof, fully detailed in Appendix D, relies on synthesizing two main observations. First, by artificially restricting the Hamiltonian to a low-energy subspace using the effective Hamiltonian approximation $\bar{H}_l = \Pi_{\leq \Delta_0} H_l \Pi_{\leq \Delta_0}$, we reduce the relevant operator norm from h down to $\|\bar{H}_l\| \leq \Delta_0$. Second, utilizing the leakage bound established in the previous section, one rigorously proves that replacing the effective evolution with the true evolution introduces only an exponentially small additional error. These two ingredients, operating in tandem, imply that the product formula behaves entirely as if the global Hamiltonian norm were merely Δ_0 instead of h.

VI. Numerical Simulations

We numerically investigate Trotter errors for quantum time evolution in two different spin systems:

- A 2×6 Heisenberg ladder (12 spins),
- A one-dimensional (1D) Heisenberg chain of 8 spins.



(a) 2×6 ladder system

(b) 1D chain (8 spins)

Fig. 1: Overview of the quantum spin lattice models.

The goal is to compare worst-case errors with errors restricted to low-energy subspaces and to analyze how these errors behave for different orders of product formulas.

A. Heisenberg Spin Model Hamiltonian

We consider a spin-$\frac{1}{2}$ Heisenberg. The Hamiltonian of the system is given by

$$H = -\sum_{\langle i,j \rangle} \left(X_i X_j + Y_i Y_j + Z_i Z_j \right), \tag{31}$$

where X_i, Y_i, Z_i are the Pauli operators acting on the i-th spin, and the summation runs over all pairs of neighboring spins $\langle i, j \rangle$.

B. Physical Interpretation

This Hamiltonian describes the isotropic Heisenberg interaction, where neighboring spins interact equally along all spatial directions. The key features of this model are:

- **Rotational symmetry:** The interaction is identical along the x, y, and z directions.
- **Ferromagnetic coupling:** The negative sign implies that the energy is minimized when spins align in the same direction.
- **Spin interaction form:** The Hamiltonian can equivalently be expressed in terms of spin operators as $H_{ij} \propto -\vec{S}_i \cdot \vec{S}_j$.

C. Relevance

This model serves as a representative example of a local many-body quantum system. Its structure allows the study of quantum dynamics, particularly the behavior of approximation errors when the evolution is restricted to low-energy states.

D. Hamiltonian Construction

In both simulations, the Hamiltonian is of the Heisenberg form:

$$H = -\sum_{\langle i,j \rangle} \left(X_i X_j + Y_i Y_j + Z_i Z_j \right), \tag{32}$$

where the summation runs over interacting pairs of spins.

1) 2×6 Ladder System: The first code constructs a ladder geometry consisting of 12 spins arranged in two rows of six sites. Interactions include:

- Horizontal nearest-neighbor couplings along each row,
- Vertical couplings connecting the two rows.

The Hamiltonian is split into two parts:

$$H = H_1 + H_2, \tag{33}$$

where interaction terms are alternated between H_1 and H_2 to ensure that each part contains mutually commuting terms. Sparse matrix representations are used for efficiency due to the large Hilbert space dimension ($2^{12} = 4096$). [5]

2) 1D Chain System: The second code constructs a one-dimensional chain of 8 spins with nearest-neighbor interactions. The Hamiltonian is similarly decomposed into:

$$H = H_1 + H_2, \tag{34}$$

where even bonds are grouped into H_1 and odd bonds into H_2. Dense matrix representations are used here since the Hilbert space is smaller ($2^8 = 256$). [5]

E. Low-Energy Subspace Construction

In both simulations, the Hamiltonian is diagonalized to obtain its eigenvalues and eigenvectors. The lowest-energy eigenstates define a sequence of subspaces:

$$\mathcal{H}_n = \text{span}\{|E_1\rangle, \ldots, |E_n\rangle\}. \tag{35}$$

Errors are evaluated for different choices of $n = 1, 50, 100, 150, 200$, allowing a systematic study of how approximation accuracy depends on the energy scale.

F. Time Evolution and Product Formulas

For a time step s, the exact evolution operator is:

$$U(s) = e^{-isH}. \tag{36}$$

This is approximated using product formulas:

1) First Order (Lie-Trotter):

$$W_1(s) = e^{-isH_1} e^{-isH_2}. \tag{37}$$

2) Second Order (Strang Splitting):

$$W_2(s) = e^{-isH_1/2} e^{-isH_2} e^{-isH_1/2}. \tag{38}$$

3) Fourth Order (Only in 1D Chain): A higher-order formula is constructed recursively using second-order steps:

$$W_4(s) = W_2(p_1 s)^2 \, W_2(p_0 s) \, W_2(p_1 s)^2, \tag{39}$$

where p_0 and p_1 are constants chosen to achieve fourth-order accuracy.

G. Error Measures

Two types of errors are computed:

1) Worst-Case Error:

$$\|U(s) - W(s)\|, \tag{40}$$

which measures the operator norm over the full Hilbert space.

2) Low-Energy Error:

$$\|(U(s) - W(s))P_n\|, \tag{41}$$

where P_n projects onto the subspace spanned by the lowest n eigenstates.

H. Results and Comparison

1) 2×6 Ladder System: The ladder system exhibits the following behavior:

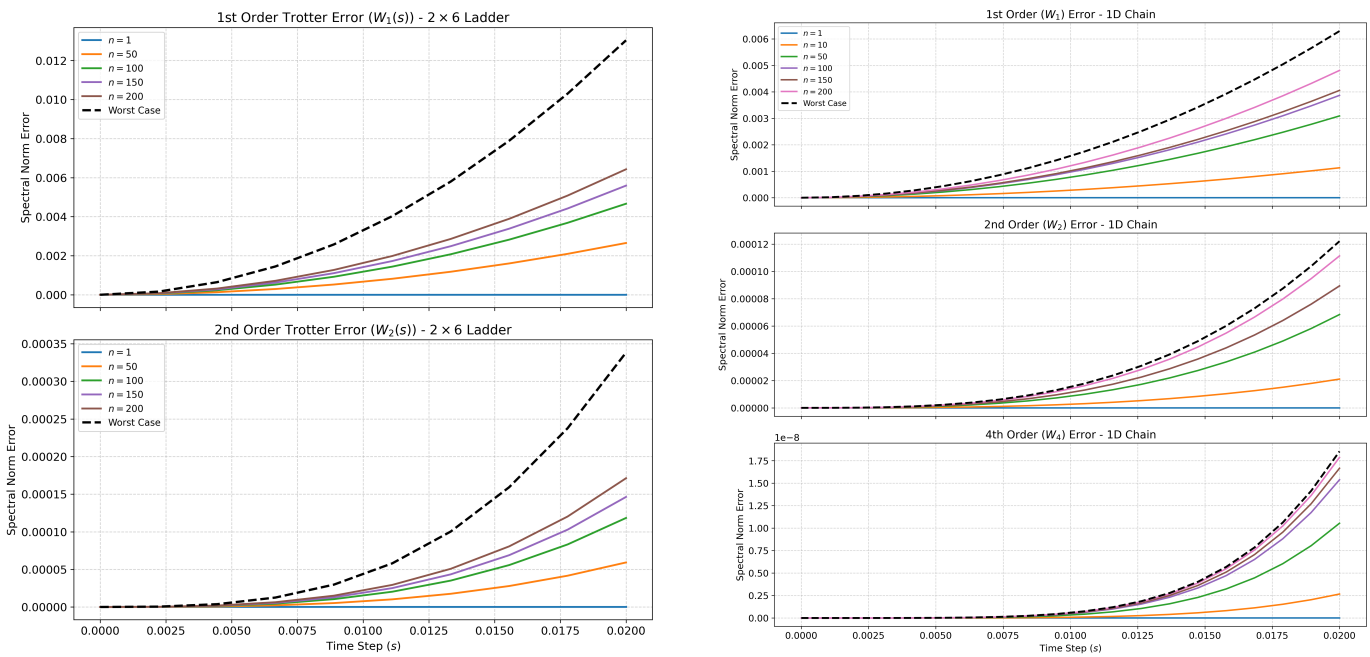
- The worst-case error grows rapidly with time step s.
- Errors restricted to low-energy subspaces are significantly smaller.
- The second-order formula reduces error substantially compared to the first-order approximation.

2) 1D Chain System: For the 1D chain:

- Similar separation between worst-case and low-energy errors is observed.
- Higher-order formulas (second and fourth order) significantly improve accuracy.
- The fourth-order method achieves very small errors even for larger time steps.

3) Key Differences:

- **Geometry:** The ladder system has higher connectivity than the 1D chain, leading to larger operator norms and generally higher errors.
- **System size:** The ladder (12 spins) is larger than the chain (8 spins), resulting in increased complexity and error accumulation.
- **Order of approximation:** The ladder study includes up to second order, while the chain includes a fourth-order method, clearly demonstrating improved scaling.
- **Error scaling:** In both systems, low-energy errors grow much more slowly than worst-case errors, confirming that worst-case bounds overestimate practical errors.



(a) 2×6 ladder system

(b) 1D chain (8 spins)

Fig. 2: Comparison of Trotter errors for different systems and product formula orders. The dashed curves represent worst-case errors, while solid curves correspond to errors restricted to low-energy subspaces. [5]

I. Conclusion

The numerical simulations demonstrate that Trotter errors are significantly reduced when restricted to low-energy subspaces. This effect is consistent across different geometries and becomes more pronounced with higher-order product formulas. These results support the theoretical insight that low-energy quantum dynamics can be simulated more efficiently than suggested by worst-case error bounds.

VII. Positivity of Local Terms and Its Physical Relevance

The error bounds derived for product formula simulations depend on the operator norm of the effective Hamiltonians $\bar{H}_l = \Pi_{\leq \Delta_0} H_l \Pi_{\leq \Delta_0}$. A key simplifying assumption is that each local term satisfies

$$H_l \geq 0, \tag{42}$$

i.e., H_l is positive semidefinite. Under this condition, for any state $|\psi\rangle$ in the low-energy subspace (with total energy bounded by Δ_0), one has

$$\langle\psi|H_l|\psi\rangle \leq \langle\psi|H|\psi\rangle \leq \Delta_0, \tag{43}$$

which implies

$$\|\bar{H}_l\| \leq \Delta_0. \tag{44}$$

This bound is crucial for obtaining improved error estimates in the low-energy regime.

In principle, the condition $H_l \geq 0$ can always be enforced by shifting each term as $H_l \to H_l + a_l\mathbb{I}$, where a_l is chosen so that the smallest eigenvalue becomes zero. However, such a shift modifies the overall energy scale of the system and can increase the effective low-energy parameter Δ (and hence Δ_0). If this increase scales with the system size N, the advantage of working in a low-energy subspace is lost, and the resulting error bounds become comparable to worst-case estimates.

Importantly, many physically relevant spin Hamiltonians naturally satisfy $H_l \geq 0$ without requiring any artificial shift. A prominent example is the Heisenberg interaction between two spins, which can be written (up to an additive constant) as

$$h_{ij} = \mathbb{I} - X_iX_j - Y_iY_j - Z_iZ_j. \tag{45}$$

Using the identity $X_iX_j + Y_iY_j + Z_iZ_j = \vec{\sigma}_i \cdot \vec{\sigma}_j$, this becomes

$$h_{ij} = \mathbb{I} - \vec{\sigma}_i \cdot \vec{\sigma}_j. \tag{46}$$

The operator $\vec{\sigma}_i \cdot \vec{\sigma}_j$ has eigenvalues $+1$ (triplet subspace) and -3 (singlet subspace), so h_{ij} has eigenvalues 0 and 4, respectively. Therefore, h_{ij} is positive semidefinite. By grouping such commuting interaction terms, each H_l can be constructed as a sum of positive semidefinite operators.

More generally, anisotropic Heisenberg models and a broad class of so-called *frustration-free Hamiltonians* also satisfy this property. A Hamiltonian $H = \sum_l H_l$ is called frustration-free if its ground state $|\psi_0\rangle$ simultaneously minimizes each local term, i.e.,

$$H_l|\psi_0\rangle = 0 \quad \forall l. \tag{47}$$

In such cases, each H_l is positive semidefinite and the ground state energy of the full Hamiltonian is exactly zero.

This structure has important consequences for low-energy simulation. Since the ground state energy is zero, low-energy states are characterized by energies $\Delta = O(1)$ that do not scale with the system size N, while the total norm of the Hamiltonian typically scales as $\|H\| = O(N)$. This separation of scales ensures that the effective norm Δ_0 remains small compared to $\|H\|$, enabling improved error bounds for product formula simulations in the low-energy subspace.

VIII. Connection Between Low-Energy Hamiltonian Simulation and Area Laws

A central theme in the study of quantum many-body systems is that physically relevant states often occupy a restricted low-energy sector of the full Hilbert space. This observation underlies both recent advances in Hamiltonian simulation and rigorous proofs of entanglement area laws. In this section, we emphasize how locality and energy constraints lead to area-law behavior.

A. Energy Localization in Local Hamiltonians

Consider a quantum spin system governed by a local Hamiltonian

$$H = \sum_X h_X, \tag{48}$$

where each term h_X acts on a finite subset of degrees of freedom. A key property of such systems is that local operators cannot efficiently generate arbitrarily large energy excitations. More precisely, if a state is supported on a low-energy subspace, then the weight of its components in high-energy sectors is exponentially suppressed.

This phenomenon appears in two complementary forms:

- **Dynamical viewpoint:** In Hamiltonian simulation restricted to low-energy states, the evolution under local terms produces only exponentially small leakage into high-energy subspaces. As a result, the dynamics can be effectively described within a truncated low-energy sector.
- **Static viewpoint:** In the analysis of energy distributions, a state with bounded global energy remains concentrated within a bounded energy window when projected onto a local Hamiltonian. Any contribution outside this window decays exponentially with the energy difference.

Both statements reflect a common principle: *local interactions constrain the redistribution of energy*, leading to an approximate decoupling between low- and high-energy sectors.

B. Effective Low-Energy Description

The suppression of high-energy components allows one to construct an effective Hamiltonian that accurately captures the behavior of low-energy states. Formally, if $\Pi_{\leq\Delta}$ denotes the projector onto energies below some threshold Δ, one can define an effective operator

$$\bar{H} = \Pi_{\leq\Delta_0} H \, \Pi_{\leq\Delta_0}, \tag{49}$$

with $\Delta_0 \geq \Delta$. The dynamics generated by $\bar{H}$ agrees with that of H on low-energy states up to small errors.

Similarly, one may define a *truncated Hamiltonian* $\tilde{H}$ in which high-energy contributions in a local region are cut off above a fixed threshold. Crucially, the low-energy spectrum and eigenstates of $\tilde{H}$ remain exponentially close to those of the original Hamiltonian. This demonstrates that low-energy physics is stable under such truncations.

C. Implications for Area Laws

The above properties have direct consequences for entanglement structure. Area laws state that, for ground states of local Hamiltonians, the entanglement entropy of a region scales with the size of its boundary rather than its volume. The connection arises through the following mechanism:

1) **Energy confinement:** Low-energy states are sharply localized in energy, with exponentially small weight in high-energy sectors.
2) **Local stability:** Truncating high-energy contributions in a spatial region changes the state only by exponentially small corrections.
3) **Bounded effective interactions:** The truncated Hamiltonian has a bounded operator norm that does not scale extensively with system size.
4) **Restricted entanglement generation:** Since interactions are local and energy fluctuations are limited, long-range entanglement cannot be generated efficiently.

These ingredients enable the approximation of ground states by tensor network states, such as matrix product states in one dimension, whose entanglement entropy obeys an area law. In particular, the truncation procedure ensures that correlations across a boundary are controlled by the strength of interactions near that boundary, leading to entropy scaling proportional to the boundary size.

D. Unified Perspective

From a broader viewpoint, both low-energy Hamiltonian simulation and area-law proofs rely on the same structural feature of local quantum systems: an effective separation between low- and high-energy sectors enforced by locality. In dynamical settings, this manifests as suppressed transitions to high-energy states during time evolution. In static settings, it appears as concentration of energy distributions and stability under local truncations.

This shared foundation highlights that area laws are not merely properties of ground states, but rather consequences of a deeper principle: *local Hamiltonians exhibit a form of locality in energy space, which constrains both dynamics and entanglement*.

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APPENDIX

A. *Proof of Error Scaling for Product Formulas*

Consider the simplest case of two Hermitian operators A and B, such that

$$H = A + B.$$

The exact time evolution operator for a short time s is

$$U(s) = e^{-is(A+B)}.$$

The first-order product formula (Lie–Trotter formula) is given by

$$W_1(s) = e^{-isA}e^{-isB}.$$

Our goal is to quantify the difference between $U(s)$ and $W_1(s)$. We begin by expanding each exponential using the Taylor series:

$$e^X = I + X + \frac{X^2}{2} + \frac{X^3}{6} + O(X^4).$$

Expanding each factor up to order s^2:

$$e^{-isA} = I - isA - \frac{s^2}{2}A^2 + O(s^3),$$

$$e^{-isB} = I - isB - \frac{s^2}{2}B^2 + O(s^3).$$

Multiplying the two expansions:

$$W_1(s) = \left(I - isA - \frac{s^2}{2}A^2\right)\left(I - isB - \frac{s^2}{2}B^2\right) + O(s^3).$$

Expanding term-by-term and retaining terms up to order s^2:

$$W_1(s) = I - is(A+B) - \frac{s^2}{2}(A^2 + B^2) - s^2AB + O(s^3).$$

Now we expand the exact operator:

$$U(s) = e^{-is(A+B)} = I - is(A+B) - \frac{s^2}{2}(A+B)^2 + O(s^3).$$

Expanding $(A+B)^2$:

$$(A+B)^2 = A^2 + B^2 + AB + BA.$$

Thus,

$$U(s) = I - is(A+B) - \frac{s^2}{2}(A^2 + B^2 + AB + BA) + O(s^3).$$

Comparing $W_1(s)$ and $U(s)$, the difference arises in the quadratic terms:

$$W_1(s) - U(s) = -s^2AB + \frac{s^2}{2}(AB + BA) + O(s^3).$$

Rewriting:

$$W_1(s) - U(s) = -\frac{s^2}{2}(AB - BA) + O(s^3).$$

Using the definition of the commutator $[A, B] = AB - BA$, we obtain:

$$W_1(s) = e^{-is(A+B) - \frac{s^2}{2}[A,B] + O(s^3)}.$$

Taking operator norms,

$$\|U(s) - W_1(s)\| = O\left(s^2\|[A,B]\|\right).$$

Using the standard inequality:

$$\|[A,B]\| \leq \|AB\| + \|BA\| \leq 2\|A\|\|B\|,$$

we obtain

$$\|U(s) - W_1(s)\| = O(h^2s^2),$$

where $h = \max(\|A\|, \|B\|)$.

Higher-order product formulas are constructed so that all BCH terms up to order s^p cancel. The remaining leading-order term appears at order s^{p+1} and consists of nested commutators of depth $p+1$. Each such term contributes a factor scaling as h^{p+1} (as shown below), leading to:

$$\|U(s) - W_p(s)\| = O(h^{p+1}|s|^{p+1}). \tag{50}$$

B. *Proof of Norm Scaling of Nested Commutators*

We now justify why nested commutators of depth k scale as $O(h^k)$. For bounded operators A and B, we use:

$$\|[A, B]\| = \|AB - BA\| \leq \|AB\| + \|BA\|.$$

Using sub-multiplicativity of operator norms, $\|AB\| \leq \|A\|\|B\|$, we obtain:

$$\|[A, B]\| \leq 2\|A\|\|B\|.$$

If $\|A\|, \|B\| \leq h$, then:

$$\|[A, B]\| = O(h^2).$$

Inductive Argument: We proceed by induction.

a) Base Case:: Depth 1: $\|[A, B]\| = O(h^2)$.

b) Inductive Step:: Assume a nested commutator of depth k satisfies

$$\|\mathcal{C}_k\| = O(h^k).$$

Now consider a depth $k+1$ commutator:

$$\mathcal{C}_{k+1} = [A, \mathcal{C}_k].$$

Then by sub-multiplicativity,

$$\|\mathcal{C}_{k+1}\| \leq 2\|A\|\|\mathcal{C}_k\| \leq 2h \cdot O(h^k) = O(h^{k+1}).$$

Thus, by induction:

$$\|\text{nested commutator of depth } k\| = O(h^k). \tag{51}$$

C. *Proof of Leakage to High-Energy States*

Let $H = \sum_l H_l$ be a local Hamiltonian acting on N spins. We define $\Pi_{\leq\Delta}$ as the projector onto energies $\leq \Delta$, and $\Pi_{>\Lambda_0}$ as the projector onto energies $> \Lambda_0$. We also let J represent the upper bound on the strength of local interaction terms, k be the maximum number of spins each local term acts on, and d be the maximum number of interaction terms involving any single spin.

We assume $\Lambda_0 \geq \Delta$. Our goal is to bound the leakage term:

$$\|\Pi_{>\Lambda_0} e^{-isH_l} \Pi_{\leq\Delta}\|.$$

We begin with the Taylor expansion:

$$e^{-isH_l} = \sum_{n=0}^{\infty} \frac{(-isH_l)^n}{n!}.$$

Applying projectors:

$$\Pi_{>\Lambda_0} e^{-isH_l} \Pi_{\leq\Delta} = \sum_{n=1}^{\infty} \frac{(-is)^n}{n!} \Pi_{>\Lambda_0} (H_l)^n \Pi_{\leq\Delta}.$$

(The $n = 0$ term vanishes since $\Pi_{>\Lambda_0}\Pi_{\leq\Delta} = 0$.)

Taking norms yields:

$$\|\Pi_{>\Lambda_0} e^{-isH_l} \Pi_{\leq\Delta}\| \leq \sum_{n=1}^{\infty} \frac{|s|^n}{n!} \|\Pi_{>\Lambda_0} (H_l)^n \Pi_{\leq\Delta}\|.$$

A key result from the locality of quantum spin systems [4] states that for any operator A,

$$\|\Pi_{>\Lambda_0} A \Pi_{\leq\Delta}\| \leq \|A\| \, e^{-\lambda(\Lambda_0 - \Delta - 2R)}, \tag{52}$$

where $\lambda = O(1/J)$ depends on locality parameters, and R is the total interaction strength of Hamiltonian terms that do not commute with A.

We apply this inequality with $A = (H_l)^n$.

Since each H_l is a sum of local terms with strength bounded by J, we have

$$\|H_l\| \leq JM,$$

where M is the number of commuting local terms in H_l.

Using submultiplicativity:

$$\|(H_l)^n\| \leq (JM)^n.$$

Each H_l acts on at most k spins. Thus, applying it n times:

$$(H_l)^n \quad \text{acts on at most } kn \text{ spins.}$$

Each spin participates in at most d interaction terms, each of strength bounded by J. Therefore, the total interaction strength of non-commuting terms satisfies:

$$R \leq Jdkn.$$

Substituting the bounds into the locality inequality:

$$\begin{aligned}\|\Pi_{>\Lambda_0}(H_l)^n\Pi_{\leq\Delta}\| &\leq (JM)^n e^{-\lambda(\Lambda_0-\Delta-2Jdkn)} \\ &= (JM)^n e^{-\lambda(\Lambda_0-\Delta)} e^{2\lambda Jdkn}.\end{aligned}$$

We combine constants into a single factor C, such that $(JM)^n e^{2\lambda Jdkn} = (CJ)^n$, giving:

$$\|\Pi_{>\Lambda_0}(H_l)^n\Pi_{\leq\Delta}\| \leq (CJ)^n e^{-\lambda(\Lambda_0-\Delta)}.$$

Substituting this back into the Taylor expansion:

$$\|\Pi_{>\Lambda_0} e^{-isH_l}\Pi_{\leq\Delta}\| \leq e^{-\lambda(\Lambda_0-\Delta)} \sum_{n=1}^{\infty} \frac{(|s|CJ)^n}{n!}.$$

Using the identity $\sum_{n=1}^{\infty} \frac{x^n}{n!} = e^x - 1$, we obtain:

$$\|\Pi_{>\Lambda_0} e^{-isH_l}\Pi_{\leq\Delta}\| \leq e^{-\lambda(\Lambda_0-\Delta)}(e^{CJ|s|} - 1). \tag{53}$$

This shows that the leakage to high-energy states is exponentially suppressed in the energy gap $\Lambda_0 - \Delta$, and the growth in leakage is strictly controlled by locality and scales with $|s|$ and system size.

D. *Proof of the Main Result*

We aim to bound the total approximation error in the low energy subspace:

$$\|(U(s) - W_p(s))\Pi_{\leq\Delta}\|.$$

To do this, we introduce an effective Hamiltonian defined by the low energy cutoff Δ_0:

$$\bar{H}_l = \Pi_{\leq\Delta_0} H_l \Pi_{\leq\Delta_0}, \quad \bar{H} = \sum_l \bar{H}_l.$$

We also define the corresponding evolution and product formula for the effective Hamiltonian:

$$\bar{U}(s) = e^{-is\bar{H}}, \quad \bar{W}_p(s).$$

We decompose the total error via the triangle inequality:

$$\begin{aligned}\|(U(s) - W_p(s))\Pi_{\leq\Delta}\| \leq &\|(U(s) - \bar{W}_p(s))\Pi_{\leq\Delta}\| \\ &+ \|(\bar{W}_p(s) - W_p(s))\Pi_{\leq\Delta}\|.\end{aligned} \tag{54}$$

Since the initial state lies completely in the low-energy subspace, the exact evolution matches the exact effective evolution:

$$U(s)\Pi_{\leq\Delta} = \bar{U}(s)\Pi_{\leq\Delta}.$$

Using standard product formula error bounds on the bounded effective operators (where $\|\bar{H}_l\| \leq \Delta_0$):

$$\|\bar{U}(s) - \bar{W}_p(s)\| = O((\Delta_0|s|)^{p+1}).$$

Thus, the first term in our error decomposition is bounded by:

$$\|(U(s) - \bar{W}_p(s))\Pi_{\leq\Delta}\| = O((\Delta_0|s|)^{p+1}).$$

Next, we bound the second term:

$$\|(\bar{W}_p(s) - W_p(s))\Pi_{\leq\Delta}\|.$$

Each exponential e^{-isH_l} in the true product formula may cause transitions outside the low-energy subspace. Using our previously derived leakage bound, these transitions satisfy:

$$\|\Pi_{>\Lambda_0} e^{-isH_l} \Pi_{\leq\Delta}\| \leq e^{-\lambda(\Lambda_0-\Delta)}(e^{CJ|s|} - 1).$$

By intelligently inserting intermediate energy projectors between the product formula time steps, one mathematically ensures that the state predominantly remains within controlled energy windows. Repeated application of the triangle inequality and the leakage bound yields:

$$\|(\bar{W}_p(s) - W_p(s))\Pi_{\leq\Delta}\| = O(\delta),$$

where δ is an error term that can be made arbitrarily small by choosing the cutoff Δ_0 to be sufficiently large.

There are two competing error contributions to balance: the product formula error $O((\Delta_0|s|)^{p+1})$, which grows with Δ_0, and the leakage error, which is exponentially suppressed in $\Delta_0 - \Delta$ and shrinks with Δ_0.

Choosing Δ_0 such that these two error sources are of comparable magnitude [2] yields the optimal scaling:

$$\Delta_0 = \Delta + O\left(J \log\left(\frac{1}{J|s|}\right) + J^2 N|s|\right). \tag{55}$$

Substituting this optimal choice back into the primary error bound completes the proof:

$$\|(U(s) - W_p(s))\Pi_{\leq\Delta}\| = O((\Delta_0|s|)^{p+1}). \tag{56}$$